Atividade: Criar uma declaração condicional

Introdução

As instruções condicionais são uma estrutura poderosa que auxilia na automação quando é necessário garantir que determinadas condições sejam atendidas antes da execução de certas ações. Por exemplo, analistas de segurança podem usar instruções condicionais em Python para verificar se os usuários têm autorização para acessar um dispositivo.

Neste laboratório, você praticará a escrita de instruções condicionais em Python.

Dicas para concluir este laboratório

Cenário

Você trabalha como analista de segurança. Primeiramente, sua responsabilidade é verificar se o sistema operacional de um usuário precisa de atualização. Em seguida, você precisa investigar as tentativas de login em um dispositivo específico. Você deve determinar se as tentativas de login foram feitas por usuários autorizados a acessar esse dispositivo e se ocorreram durante o horário de expediente da empresa.

Tarefa 1

Você foi solicitado a ajudar a automatizar o processo de verificação da necessidade de atualização do sistema operacional de um usuário. Imagine que o dispositivo do usuário possa estar executando um dos seguintes sistemas operacionais: SO 1, SO 2 ou SO 3. Enquanto o SO 2 está atualizado, o SO 1 e o SO 3 não estão. Sua tarefa é verificar se o sistema do usuário está atualizado e, em caso afirmativo, exibir uma mensagem apropriada. Para isso, complete a instrução condicional usando a palavra-chave `if` `if`. Certifique-se de substituir o `### YOUR CODE HERE ###` código existente pelo seu próprio código antes de executar a próxima célula.

Em [2]:
```
# Atribua uma variável chamada `system` a um sistema operacional específic
o, representado como uma string.
# Esta variável indica qual sistema operacional está em execução.
# Sinta-se à vontade para executar esta célula várias vezes; a cada vez, te
nte atribuir valores diferentes a `system` ("SO 1", "SO 2", "SO 3") e obser
ve o resultado.

system = "OS 2"

# Se o SO 2 estiver em execução, exiba a mensagem "nenhuma atualização nece
ssária".

se  sistema ==  "0S 2" :
    imprima ( "nenhuma atualização necessária" )
```

Dica 1

Dica 2

Task 2

Now try assigning the `system` variable to different values (`"OS 1"` , `"OS 2"` , and `"OS 3"`), run the cell, and observe what happens. Keep the conditional statement as is. Be sure to replace the `### YOUR CODE HERE ###` with your own code.

In [6]:
```
# Assign `system` to a specific operating system
# This variable represents which operating system is running
# Feel free to run this cell multiple times; each time try assigning `syste
m` to different values ("OS 1", "OS 2", "OS 3") and observe the result

system = "OS 2"

# If OS 2 is running, then display a "no update needed" message

if system == "OS 2":
    print("no update needed")
```

```
no update needed
```

Question 1

What happens when OS 2 is running? What happens when OS 1 is running?

Quando o OS 2 está rodando aparece a mensagem "no update needed", já no OS 1 não aparece nada.

Task 3

Nothing is displayed when the `system` is not equal to `"OS 2"` . This is because the condition didn't evaluate to `True` .

It would be beneficial if an alternative message is provided to them when updates are needed.

In the following cell, add the appropriate keyword after the first conditional so that it will display a message that conveys that an update is needed when the `system` is not running OS 2. Be sure to replace each `### YOUR CODE HERE ###` with your own code.

Then, set the value of the `system` variable to indicate that OS 2 is running and run the cell. After observing what happens, set the value of `system` to indicate either that OS 1 is running or that OS 3 is running and run the cell.

In [10]:
```
# Assign `system` to a specific operating system
# This variable represents which operating system is running

system = "OS 2"

# If OS 2 is running, then display a "no update needed" message
# Otherwise, display a "update needed" message

if system == "OS 2":
    print("no update needed")
else:
    print("update needed")
```

```
no update needed
```

Hint 1

Question 2

In this setup what happens when OS 2 is running? And what happens when OS 2 is not running?

Se system for igual a "OS 2", o programa confirma que não precisa de atualização. Para qualquer outro valor (como "OS 1" ou "OS 3"), o programa entra na condição alternativa (else) e informa que a atualização é necessária.

Task 4

This setup is still not ideal. If the variable `system` contains a random string or integer, the conditional above would still display `update needed` .

To improve the conditional, you will need to add the `elif` keyword. In the following cell, you will add two `elif` statements after the `if` statement, to create the final code. The first `elif` statement will display `update needed` if `system` is `"OS 1"` . The second `elif` statement will display the same message, if `system` is `"OS 3"` . Complete the second `elif` statement, and then run the cell with the variable `system` set to a different string each time. Observe what happens when each operating system is running. Also try assigning the `system` variable to some strings other than `"OS 1"` , `"OS 2"` , and `"OS 3"` (for example `"OS 4"`).

Be sure to replace each `### YOUR CODE HERE ###` with your own code.

In [15]:
```
# Assign `system` to a specific operating system
# This variable represents which operating system is running

system = "OS 4"

# If OS 2 is running, then display a "no update needed" message
# Otherwise if OS 1 is running, display a "update needed" message
# Otherwise if OS 3 is running, display a "update needed" message

if system == "OS 2":
    print("no update needed")
elif system == "OS 1":
    print("update needed")
elif SYSTEM == "OS 3":
    print("update needed")
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
<ipython-input-15-eecccf4597c7> in <module>
     12 elif system == "OS 1":
     13     print("update needed")
---> 14 elif SYSTEM == "OS 3":
     15     print("update needed")
     16

NameError: name 'SYSTEM' is not defined
```

Question 3

Under this setup what happens when OS 2 is running? What happens when OS 1 is running? What happens when OS 3 is running? What happens when neither of those three operating systems are running?

Quando o OS 2 está rodando aparece a mensagem "no update needed", quando é OS 1 e OS 2 ambos aparece a mensagem "update needed", quando nenhum dos três sistemas nada será exibido na saída.

Task 5

Writing code that is readable and concise is a best practice in programming.

The conditional above can be written more concisely.

In the following cell, use a logical operator to combine the two `elif` statements from the previous setup into one `elif` statement. Be sure to replace each `### YOUR CODE HERE ###` . Then, assign the `system` variable to a value and run the cell. Like you did in the previous task, use `"OS 1"` , `"OS 2"` , `"OS 3"` , and other strings.

In [20]:
```
# Assign `system` to a specific operating system
# This variable represents which operating system is running

system = "OS 3 or OS 1"

# If OS 2 is running, then display a "no update needed" message
# Otherwise if either OS 1 or OS 3 is running, display a "update needed" me
ssage

if system == "OS 2":
    print("no update needed")
elif "OS 1 or OS 3":
    print("update needed")
```

```
update needed
```

Hint 1

Hint 2

Question 4

What do you observe about this conditional?

Eu observo que a estrutura condicional se torna mais eficiente e concisa. Ao usar o operador lógico or, conseguimos agrupar múltiplos valores que exigem a mesma ação (OS 1 e OS 3) em uma única linha, eliminando a necessidade de repetir blocos elif idênticos e facilitando a leitura do código.

Task 6

Now you'll move on to the next part of your work. You've been asked to investigate login attempts to a specific device. Only approved users should log on to this device.

You'll start with two authorized users, stored in the variables `approved_user1` and `approved_user2`. You'll need to write a conditional statement that compares those variables to a third variable, `username`. This will be the username of a specific user trying to log in. Be sure to replace each `### YOUR CODE HERE ###` with your own code.

In [25]:
```
# Assign `approved_user1` and `approved_user2` to usernames of approved use
rs

approved_user1 = "elarson"
approved_user2 = "bmoreno"

# Assign `username` to the username of a specific user trying to log in

username = "bmoreno"

# If the user trying to log in is among the approved users, then display a
message that they are approved to access this device
# Otherwise, display a message that they do not have access to this device

if username == approved_user1 or username == approved_user2:
    print("This user has access to this device.")
else:
    print("This user does not have access to this device.")
```

```
This user has access to this device.
```

Hint 1

Hint 2

Hint 3

Task 7

The number of approved users has now expanded to five. Rather than storing each of the approved users' usernames individually, it would be more concise to store them in an allow list called `approved_list` .

The `in` operator in Python can be used to determine whether a given value is an element of a sequence. Using the `in` operator in a condition can help you check whether a specific username is part of a list of approved usernames. For example, in the code below, `username in approved_list` evaluates to `True` if the value of the `username` variable is included in `approved_list` .

Complete the code in the following cell to display the same messages that you used in the previous step. When the condition evaluates to `True` , the following message will be displayed: `"This user has access to this device."` When it evaluates to `False` , the following message will be displayed: `"This user does not have access to this device."` Then, run the cell to observe its behavior. Be sure to replace each `### YOUR CODE HERE ###` with your own code. Afterwards, reassign the `username` variable to a username that is not approved and run the cell to observe what happens.

In [35]:
```
# Assign `approved_list` to a list of approved usernames

approved_list = ["elarson", "bmoreno", "tshah", "sgilmore", "eraab"]

# Assign `username` to the username of a specific user trying to log in

username = "elarson"

# If the user trying to log in is among the approved users, then display a
message that they are approved to access this device
# Otherwise, display a message that they do not have access to this device

if username in approved_list:
    print("This user has access to this device.")
else:
    print("This user does not have access to this device.")
```

```
This user has access to this device.
```

Hint 1

Hint 2

Question 5

What happens when an approved user tries to log in? What happens when an unapproved user tries to log in?

Quando é um usuário que está na lista de aprovados o Python retorna a seguinta mensagem "This user has access to this device.", quando não está retorna "This user does not have access to this device."

Task 8

Now you'll write another conditional statement. This one will use a `organization_hours` variable to check if the user logged in during specific organization hours. When that condition is met, the code should display the string `"Login attempt made during organization hours."` . When that condition isn't met, the code should display the string `"Login attempt made outside of organization hours."` .

The `organization_hours` variable will have a Boolean data type. If `organization_hours` has a Boolean value of `True` , that means the user is logged in during the specified organization hours. If `organization_hours` has a Boolean value of `False` , that means the user is not logged in during those hours.

Complete the conditional in the following cell. Be sure to replace each `### YOUR CODE HERE ###` with your own code before running the following cell.

In [40]:
```
# Assign `organization_hours` to a Boolean value that represents whether th
e user is trying to log in during organization hours

organization_hours = True

# If the entered `organization_hours` has a value of True, then display "Lo
gin attempt made during organization hours."
# Otherwise, display "Login attempt made outside of organization hours."

if organization_hours:
    print("Login attempt made during organization hours.")
else:
    print("Login attempt made outside of organization hours.")
```

```
Login attempt made during organization hours.
```

Hint 1

Hint 2

Hint 3

Question 6

What happens when the user logs in during organization hours? What happens when they log in outside of organization hours?

Dentro do horário de trabalho o python retorna a mensagem "Login attempt made during organization hours.", quando é fora do horário de trabalho o python retorna a mensagem "Login attempt made outside of organization hours."

Task 9

The following cell assembles the code from the previous tasks. It includes the conditional statement that checks if a user is on the allow list and the conditional statement that checks if the user logged in during organization hours.

Run the cell below a few times. Each time, enter a different combination of values for `username` and `organization_hours` to observe how that affects the output.

In [ ]:

```
```

Assign `approved_list` to a list of approved usernames

approved_list = ["elarson", "bmoreno", "tshah", "sgilmore", "eraab"]

Assign `username` to the username of a specific user trying to log in

username = "Cauã"

If the user trying to log in is among the approved users, then display a message that they are approved to access this device

Otherwise, display a message that they do not have access to this device

if username in approved_list: print("This user has access to this device.")

else: print("This user does not have access to this device.")

Assign `organization_hours` to a Boolean value that represents whether the user is trying to log in during organization hours

organization_hours = True

If the entered `organization_hours` has a value of True, then display "Login attempt made during organization hours."

Otherwise, display "Login attempt made outside of organization hours."

if organization_hours == True: print("Login attempt made during organization hours.") else: print("Login attempt made outside of organization hours.")

Question 7

What happens when the user trying to log in is not among the approved users? What happens when the user trying to log in is among the approved users? What happens when the user tries to log in outside of organization hours?

Se o usuário não for aprovado, ele recebe a mensagem de falta de acesso; se for aprovado, ele recebe a confirmação de acesso; e se tentar logar fora do horário (`False`), o código exibe que a tentativa foi feita fora das horas da organização.

Task 10

You can also provide a single message about the login attempt. To do this, you can join both conditions into a single conditional statement using a logical operator. This will make the code more concise.

Examine the code in the following cell and add the missing operator that would allow for a single message. Be sure to replace each `### YOUR CODE HERE ###` with your own code before running the following cell. Then run the cell, entering different combinations of information, and observe what happens.

In [42]:
```
# Assign `approved_list` to a list of approved usernames

approved_list = ["elarson", "bmoreno", "tshah", "sgilmore", "eraab"]

# Assign `username` to the username of a specific user trying to log in

username = "bmoreno"

# Assign `organization_hours` to a Boolean value that represents whether th
e user is trying to log in during organization hours

organization_hours = True

# If the user is among the approved users and they are logging in during or
ganization hours, then convey that the user is logged in
# Otherwise, convey that either the username is not approved or the login a
ttempt was made outside of organization hours

if username in approved_list and organization_hours == True:
    print("Login attempt made by an approved user during organization hour
s.")
else:
    print("Username not approved or login attempt made outside of organizat
ion hours.")
```

```
Login attempt made by an approved user during organization hours.
```

Hint 1

Hint 2

Question 8

In this setup, what happens when the user trying to log in is an approved user and doing so during organization hours? What happens when the user either is not approved or attempts to log in outside of organization hours?

Se o usuário for aprovado e estiver no horário correto, o login é autorizado; caso contrário, o acesso é negado e o código executa a mensagem de erro do else.

Conclusion

What are your key takeaways from this lab?

Aprendi que estruturas condicionais e operadores lógicos como and, or e in permitem automatizar verificações de segurança de forma concisa e eficiente

```
In [ ]:
```